Huawei Access Node (YEL-ACC-22) QoS Congestion Event Remediation SOP

Node: YEL-ACC-22 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_8679

Description

This Standard Operating Procedure (SOP) addresses QoS congestion events occurring on the Huawei Access Node YEL-ACC-22 at the Yelahanka Sub-station, characterized by Microbursts, interface tail drops, and QoS queue starvation. The Huawei Access Node, as a critical component in the network infrastructure, employs QoS mechanisms to prioritize and manage traffic flow. When QoS congestion events occur, it may be due to the misconfiguration of QoS rules, over-subscription of interfaces, or excessive traffic bursts. The Huawei Access Node utilizes the QoS framework, which is built on the concept of queuing disciplines (QDiscs) to manage traffic flow. Each QDisc applies specific policies to packets, enabling the node to prioritize and regulate traffic based on predefined criteria. In the event of a congestion, the Huawei QDisc may enter a tail-drop state, leading to packet loss and degradation of QoS. This SOP provides a structured approach to diagnosing and resolving QoS congestion events on the Huawei Access Node YEL-ACC-22.

Pre-requisite Checks

- Remote Hands dispatch to the Yelahanka Sub-station for physical access.
- Verify the status of redundant links and ensure they are operational.
- Confirm the Huawei Access Node YEL-ACC-22 is powered on and has a stable system clock.

Potential Root Causes

Cause	Probability
Misconfigured QoS rules leading to over-subscription of interfaces.	High
Excessive traffic bursts causing QoS queue starvation.	Medium
Huawei QDisc misconfiguration resulting in tail-drop state.	Low

Diagnostic Commands & Expected Outputs

```
Command: display qce queue-stat, Expected Output: Queue statistics for each QDisc, including packet count, byte count, and drop count., Description: This command displays the statistics for each QDisc on the Huawei Access Node, enabling analysis of traffic flow and identifying potential congestion points.

Command: display qce policy-stat, Expected Output: Policy statistics for each QDisc, including policy hit count and drop count., Description: This command displays the policy statistics for each QDisc, allowing for the identification of specific policies causing congestion.
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Command: display interface brief, Expected Output: Interface statistics, including ingress and egress packet and byte count., Description: This command displays the interface statistics, enabling the identification of excessive traffic bursts or over-subscription of interfaces.
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Resolution Steps

Step 1: Step: Verify and correct QoS rule configurations., Command: undo qce rule , Description: This command removes the specified QoS rule, allowing for the verification of its impact on traffic flow and congestion.

Step 2: Step: Adjust QoS queue configurations to prevent over-subscription., Command: qce queue size , Description: This command adjusts the size of the specified QoS queue to prevent over-subscription and alleviate congestion.

Step 3: Step: Optimize Huawei QDisc configurations to prevent tail-drop state., Command: qce discard-mode , Description: This command optimizes the Huawei QDisc configuration to prevent the tail-drop state, ensuring QoS queue stability.

Post-Check Verification

```
Command: display qce queue-stat, Expected Output: Queue statistics for each QDisc, indicating stable traffic flow., Description: This command verifies the stability of traffic flow and QoS queue statistics after implementing the resolution steps.

Command: display interface brief, Expected Output: Interface statistics, indicating no excessive traffic bursts or over-subscription., Description: This command verifies the absence of excessive traffic bursts or over-subscription of interfaces after implementing the resolution steps.
```

Roll-back Procedures

- Step: Restore original QoS rule configurations., Command: qce rule , Description: This command restores the original QoS rule configuration in case the fix fails.

- Step: Undo QoS queue adjustments., Command: undo qce queue size , Description: This command reverts the QoS queue size adjustment in case the fix fails.

- Step: Revert Huawei QDisc configurations., Command: undo qce discard-mode , Description: This command reverts the Huawei QDisc configuration in case the fix fails.